Range Migration Algorithm for Near-Field MIMO-SAR Imaging

Rongqiang Zhu, Jianxiong Zhou, Ge Jiang, and Qiang Fu

***Abstract*— In this letter, a 3-D range migration algorithm for multiple-input multiple-output synthetic aperture radar imaging is proposed. The accurate expression of the signal spectrum is derived by utilizing the spherical wave decomposition. This method compensates for the curvature of the wavefront in wavenumber domain and achieves the 3-D Fourier transform of the reflectivity map through a dimension-reducing accumulation operation. Its fast Fourier transform-based imaging scheme provides high imaging efficiency. In addition, this method does not take the plane wave approximation and can be applied in near-field imaging scenes. Both theoretical analysis and experimental results show that the proposed method can reconstruct the 3-D image as accurately as the backprojection algorithm but with much lower computational load.**

***Index Terms*— Multiple-input multiple-output synthetic aperture radar (MIMO-SAR), multistatic configuration, near-field imaging, range migration algorithm (RMA).**

I. Introduction

SYNTHETIC aperture radar (SAR) has a wide range of applications in remote sensing due to its unique advantages. However, the imaging result obtained by SAR is the projection of the 3-D reflectivity map onto an imaging plane. To overcome this shortcoming, a novel imaging system combining linear array and synthetic aperture technology for 3-D imaging is proposed [1]–[3]. This method synthesizes a 2-D array by scanning a multiple-input multiple-output (MIMO) array in the direction perpendicular to the array. This structure takes advantage of wide illumination and short data acquisition time of the MIMO array [4]. However, compared with monostatic array, its imaging method is more complicated due to separated transmitting and receiving elements.

In SAR, the range migration algorithm (RMA) is one of the most efficient imaging methods and has been widely used. It provides high imaging efficiency by the application of the fast Fourier transform (FFT) and can be used under the near-field condition [5]. Sheen *et al.* [6] extended the RMA for near-field 3-D radar imaging. Nevertheless, the RMA cannot be directly applied to MIMO array. In order to extend the RMA to multistatic configurations, Moulder *et al.* [7] adjusted the echoed data into a monostatic format according to the equivalent phase center principle and compensated for the adjusted data by a reference signal; Álvarez *et al.* [8] introduced a Fourier-based method using multiple-plane-wave approximation and wavenumber space partitioning; and Du *et al.* [9] derived a method for single-input multiple-output (SIMO)-SAR by expanding the phase term to its third-order Taylor series. The approximations used in these methods either constrain the application scene or reduce the imaging precision. A novel frequency-domain imaging method for a SIMO array is presented in [10]. This method avoids interpolation by utilizing the FFT-based subimaging scheme; however, it cannot be extended to MIMO-SAR. Zhuge and Yarovoy [11] proposed an extended RMA suitable for MIMO array when both the receiving and the transmitting arrays satisfy the Nyquist criterion. But this requirement is generally unsatisfied in MIMO arrays.

The backprojection algorithm (BPA) is another imaging method commonly used in SAR. This method provides high imaging precision by coherently accumulating the received signal of each antenna element and can be used for arbitrary array configurations. But its efficiency is too low for 3-D MIMO-SAR imaging.

In this letter, we propose an efficient frequency-domain imaging algorithm for MIMO-SAR. Similar to the traditional RMA, this method transforms the measurements into the wavenumber domain for compensation and interpolation. Then, an accumulation operation is performed to transform the wavenumber domain data from the 4-D space into a 3-D space. Finally, the imaging result is obtained by 3-D inverse FFT. Compared with BPA, this method can achieve similar imaging precision but greatly reduce the computational load.

II. Formulation

The imaging geometry of MIMO-SAR is shown in Fig. 1. The Cartesian coordinate system O–xyz is established; the origin is located at the center of a target; and x-, y-, and z-axes denote the azimuth, range, and height directions, respectively. A linear MIMO array consisting of N_R receiving elements and N_T transmitting elements illuminates a target at a distance y_0 and scans along the z-axis. The location of transmitting elements is denoted by (u, y_0, h), and the location of receiving elements is denoted by (v, y_0, h). $\mathbf{P}$ represents an arbitrary point on the target with location (x, y, z) and reflectivity $\sigma(x, y, z)$. R_t is the distance between the transmitting element and point $\mathbf{P}$, and R_r is the distance between the receiving element and point $\mathbf{P}$. R_t and R_r are given by

$$R_t = \sqrt{(u-x)^2 + (y_0-y)^2 + (h-z)^2} \tag{1}$$

$$R_r = \sqrt{(v-x)^2 + (y_0-y)^2 + (h-z)^2}. \tag{2}$$

Manuscript received August 22, 2017; accepted October 4, 2017. Date of publication November 2, 2017; date of current version December 4, 2017. This work was supported in part by NSAF under Grant U1630101 and in part by the National Natural Science Foundation of China under Grant 61601424. *(Corresponding author: Jianxiong Zhou.)*

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Digital Object Identifier 10.1109/LGRS.2017.2761838

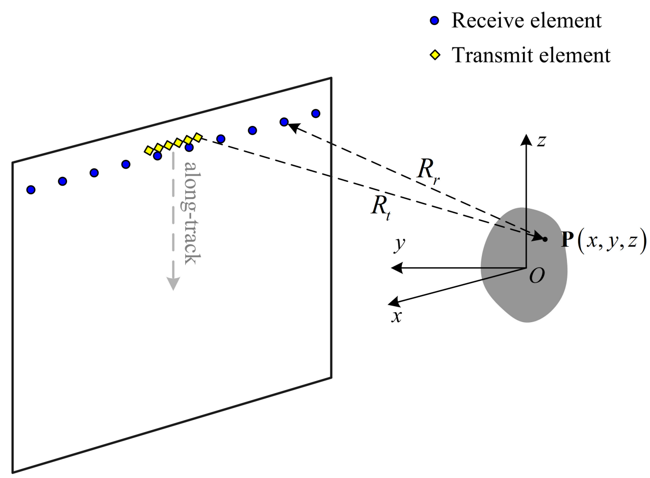

Fig. 1. Imaging geometry of MIMO-SAR.

Assuming that the transmitted signal is a stepped-frequency continuous wave, the received signal at the corresponding transceiver pair can be expressed as

$$s(u, \upsilon, k, h) = \iiint_{\Omega(x,y,z)} \sigma(x, y, z) e^{-jkR_r} e^{-jkR_t} dxdydz \tag{3}$$

where $\Omega(x, y, z)$ represents the distribution space of a distributed target, and $k = 2\pi f/c$ is the wavenumber, with c denoting the speed of light and f representing the frequency.

According to the imaging principle of time domain correlation algorithm [5], the reflectivity map can be estimated by

$$\hat{\sigma}(x, y, z) = \int\int\int\int s(u, \upsilon, k, h) e^{jkR_r} e^{jkR_t} dud\upsilon dkdh \tag{4}$$

where the exponential term represents a spherical wave propagation model. In the Appendix, we derive the spherical wave decomposition equation. By applying the variable replacement to (A5), the exponential term in (4) can be decomposed as

$$e^{jkR_t} = \int_{\Omega(k_u)} e^{-jk_u(u-x)} e^{j\sqrt{k^2-k_u^2}\sqrt{(y_0-y)^2+(h-z)^2}} dk_u \tag{5}$$

$$e^{jkR_r} = \int_{\Omega(k_v)} e^{-jk_v(\upsilon-x)} e^{j\sqrt{k^2-k_v^2}\sqrt{(y_0-y)^2+(h-z)^2}} dk_v \tag{6}$$

where $\Omega(k_u)$ and $\Omega(k_v)$ represent the distribution space of k_u and k_v, respectively, determined by

$$k_u = k\frac{x-u}{\sqrt{(u-x)^2+(y_0-y)^2+(h-z)^2}} \tag{7}$$

$$k_v = k\frac{x-\upsilon}{\sqrt{(\upsilon-x)^2+(y_0-y)^2+(h-z)^2}}. \tag{8}$$

Inserting (5) and (6) into (4) yields

$$\hat{\sigma}(x, y, z) = \iiiint S'(k_u, k_v, k, h) e^{j(k_u+k_v)x} \times e^{j\left(\sqrt{k^2-k_u^2}+\sqrt{k^2-k_v^2}\right)\sqrt{(y_0-y)^2+(h-z)^2}} dk_u dk_v dkdh \tag{9}$$

where $S'(k_u, k_v, k, h)$ is the Fourier transform of the received signal on u and υ, and k_u and k_v represent the Fourier transform variables corresponding to u and υ, respectively. The second exponential term in (9) indicates a spherical wave propagation model in the 2-D space O–yz (the space perpendicular to the MIMO array), which is similar to the spherical wave propagation model in the monostatic case. By substituting $k' = (k^2-k_u^2)^{1/2} + (k^2-k_v^2)^{1/2}$, $u = h$, $x = z$, and $k_u = k_z$ into (A5), the following relationship can be obtained:

$$e^{j\left(\sqrt{k^2-k_u^2}+\sqrt{k^2-k_v^2}\right)\sqrt{(y_0-y)^2+(h-z)^2}} = \int e^{-jk_z(h-z)} e^{-jk_y(y_0-y)} dk_z \tag{10}$$

where

$$k_y = \sqrt{\left(\sqrt{k^2-k_u^2}+\sqrt{k^2-k_v^2}\right)^2 - k_z^2}. \tag{11}$$

By using (10), (9) can be simplified as

$$\hat{\sigma}(x, y, z) = \iiiint S(k_u, k_v, k, k_z) \times e^{j(k_u+k_v)x} e^{-jk_y(y_0-y)} e^{jk_z z} dk_u dk_v dkdk_z \tag{12}$$

where $S(k_u, k_v, k, k_z)$ is the Fourier transform of $S'(k_u, k_v, k, h)$ on h and represents the wavenumber domain echoed data; k_z represents the Fourier transform variable corresponding to h. The exponential term $e^{-jk_y y_0}$ can be compensated for by multiplying in wavenumber domain, i.e.,

$$\tilde{S}(k_u, k_v, k, k_z) = S(k_u, k_v, k, k_z) e^{-jk_y y_0}. \tag{13}$$

After compensation, in order to directly perform FFT for range imaging, the Stolt interpolation is utilized to resample the data in domain k_y according to (11). The resampled signal is labeled by $\tilde{S}_r(k_u, k_v, k_y, k_z)$. The relationship between the resampled data and the reflectivity is as follows:

$$\hat{\sigma}(x, y, z) = \iiiint \tilde{S}_r(k_u, k_v, k_y, k_z) \times e^{j(k_u+k_v)x} e^{jk_y y} e^{jk_z z} dk_u dk_v dk_y dk_z. \tag{14}$$

The wavenumber domain echoed data are distributed in 4-D space $k_u - k_v - k_y - k_z$, whereas the reflectivity is distributed in a 3-D space. Clearly, the reflectivity is not directly obtained by Fourier transform of the wavenumber domain data. Therefore, the dimension-reducing accumulation operation is utilized to transform the 4-D data into a 3-D format. This operation accumulates the data distributed in space k_u-k_v into a common space k_x according to the following relation:

$$k_x = k_u + k_v. \tag{15}$$

The accumulated signal is labeled by $F(k_x, k_y, k_z)$. The data before and after dimension-reducing accumulation have the following relationship:

$$F(k_x, k_y, k_z) = \iint_{k_x=k_u+k_v} \tilde{S}_r(k_u, k_v, k_y, k_z) dk_u dk_v. \tag{16}$$

Using (16), (14) can be rewritten as

$$\hat{\sigma}(x, y, z) = \iiint F(k_x, k_y, k_z) e^{jk_x x} e^{jk_y y} e^{jk_z z} dk_x dk_y dk_z. \tag{17}$$

It can be clearly seen from (17) that the reflectivity is the 3-D inverse Fourier transform of $F(k_x, k_y, k_z)$.

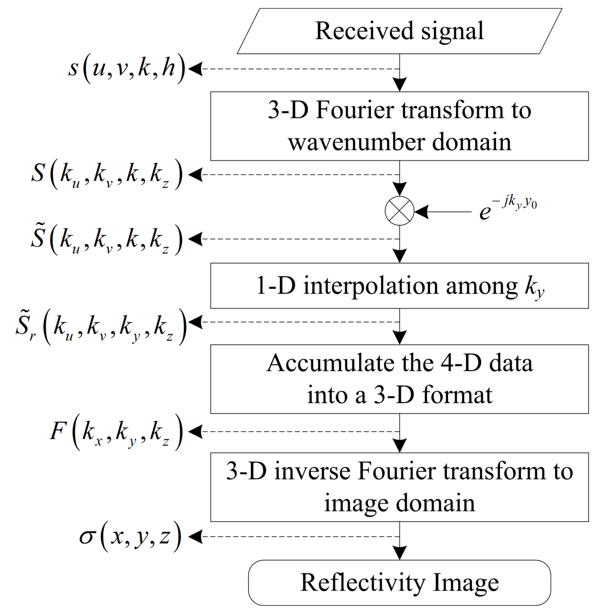

Fig. 2. Flowchart of the proposed method.

III. Algorithm Implementation and Computational Load Analysis

The flowchart of the proposed method is illustrated in Fig. 2. Its procedure can be decomposed into the following four steps.

Step 1: Transform the received signal into the wavenumber domain and compensate for the phase shift. In order to improve the efficiency, for uniformly spaced receiving and transmitting elements, this transformation can be efficiently implemented by 3-D FFT. However, for general MIMO array, the spatial sampling of the receiving or transmitting array does not satisfy the Nyquist criterion. The spectrum data obtained by FFT are overlapped due to undersampling, and the space of the spectrum is part of $\Omega(k_u)$ and $\Omega(k_v)$. In this case, to meet the condition of (5) and (6), the echoed data should be upsampled before FFT, or the spectrum data should be replicated after FFT. The overlapped spectrum data result in grating lobes of imaging results. The grating lobes are caused by undersampling rather than by the imaging method; therefore, suppressing grating lobes is not considered in this letter.

The computational load is measured by the floating point operation (FLOP) required by the method. Each FLOP can be either a real multiply or a real add [12]. The computational loads of basic operations are listed in Table I. In this step, there is a 3-D FFT and $N_u N_v N_z N_k$ complex multiplies. The 3-D FFT can be decomposed into three 1-D FFTs. Therefore, the computational load for this step is

$$C_1 \approx \underbrace{5N_u N_v N_z N_k \log_2(N_u N_v N_z)}_{\text{3-D FFT}} + \underbrace{6N_u N_v N_z N_k}_{\text{Compensation}} \text{ FLOP} \tag{18}$$

where N_u, N_v, and N_z are the points of FFT in receiving, transmitting, and height directions, respectively, and N_k is the number of frequency samples. For the sake of simplicity, all the points of FFT are assumed to be a power of 2, and a radix-2 FFT is taken.

TABLE I

Computational Load of Basic Operations (FLOP)

Complex add	Complex multiply	Radix-2 FFT/IFFT [a]	Interpolation [b]
2	6	$5N\log_2(N)$	$2(2N_{\text{ker}}-1)$

[a] N is the point of FFT/IFFT; [b] N_{ker} is the length of interpolation kernel function.

Step 2: Resample the wavenumber domain data in domain k_y. The resampling can be implemented by interpolation. In this letter, we use linear interpolation kernel function, whose length is 2. From Table I, we can calculate that the computational load required per output point is six FLOPs. The computational load for this step is

$$C_2 \approx 6N_u N_v N_y N_z \text{ FLOP} \tag{19}$$

where N_y is the length of the resampled data in domain k_y.

Step 3: Accumulate the resampled data from the 4-D space k_u-k_v-k_y-k_z into the 3-D space k_x-k_y-k_z. The accumulated data are evenly distributed in domain k_x. The support along k_x is determined by $\Omega(k_u)$ and $\Omega(k_v)$, and the discrete interval is determined by the intervals of k_u and k_v. This dimension-reducing accumulation procedure makes it possible to directly achieve the reflectivity by FFT. To facilitate the accumulation, the sampling interval of k_u and k_v should be the same. Therefore, zero padding is performed prior to the FFT operation to produce receiving and transmitting arrays of equal length. There are $(N_u N_v - N_x)N_y N_z$ complex adds in this step. The computational load is

$$C_3 \approx 2(N_u N_v - N_x)N_y N_z \text{ FLOP} \tag{20}$$

where N_x is the length of the accumulated data in domain k_x.

Step 4: Transform the data from the wavenumber domain into image domain by 3-D inverse FFT. Similar to the analysis in Step 1, the computational load for this step is

$$C_4 \approx 5N_x N_y N_z \log_2(N_x N_y N_z) \text{ FLOP.} \tag{21}$$

In summary, the total computational load of the proposed method is

$$\begin{aligned} C_p \approx\ & 5N_u N_v N_z[N_k \log_2(N_u N_v N_z) + 1.2N_k + 1.6N_y] \\ & + 5N_x N_y N_z[\log_2(N_x N_y N_z) - 0.4] \text{ FLOP.} \end{aligned} \tag{22}$$

Under the same imaging conditions, the computational load of BPA is [11]

$$C_{\text{BPA}} \approx 8N_T N_R N_h N_x N_y N_z \text{ FLOP} \tag{23}$$

where N_h is the number of height samples.

From (21) and (23), it can be seen that the proposed method greatly reduces the computational load compared with the BPA.

IV. Experimental Validation

The proposed imaging method has been validated by both simulated data and real data. The imaging geometry is the same as that shown in Fig. 1. The imaging methods are implemented in MATLAB using a computer with a CPU at 2.8 GHz and 8-GB RAM.

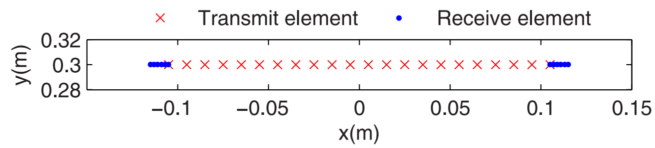

Fig. 3. Topology of the MIMO array used in the simulation.

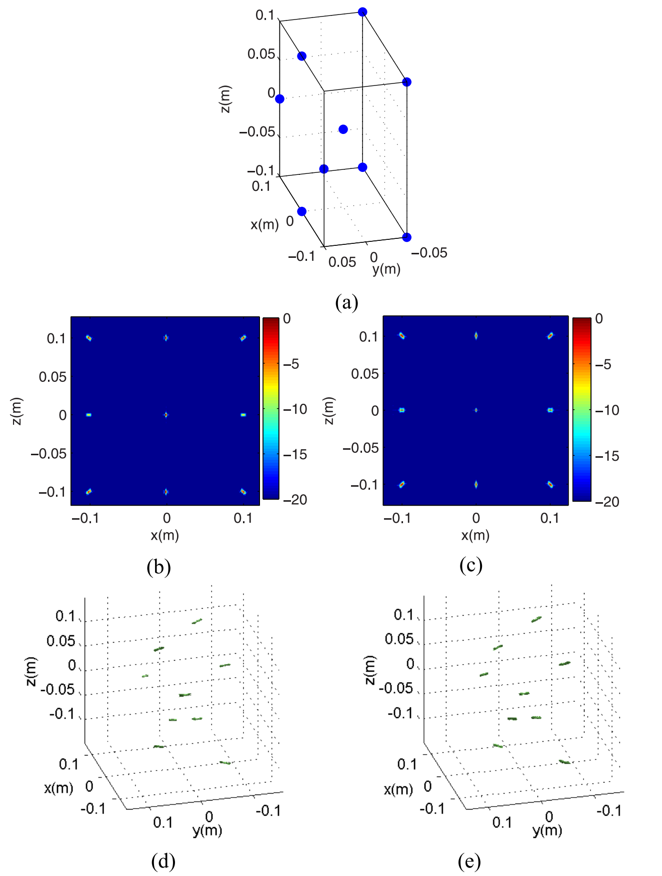

Fig. 4. Imaging results of simulation data. (a) Measurement setup. (b) Front view of image obtained by the proposed method. (c) Front view of image obtained by BPA. (d) Three-dimensional image obtained by the proposed method. (e) Three-dimensional image obtained by BPA. The front view is obtained by maximum projection of the 3-D image onto the *xz* plane.

A. Numerical Simulations

Fig. 3 shows the topology of the MIMO array used in simulation. The MIMO array consists of a uniform sampling transmitting array and two uniform sampling receiving subarrays. The transmitting array consists of 22 elements with intervals of 10 mm. The receiving subarray consists of six elements with intervals of 2 mm. The distance between two receiving subarrays is 210 mm. All transmitting elements and receiving elements are located at 0.3 m from the target center along the x-axis. The length of the synthetic aperture is 256 mm with sampling intervals of 1 mm. The target consists of nine ideal point scatterers as shown in Fig. 4(a). The working frequency is 130–150 GHz with a frequency step 100 MHz.

The 3-D reflectivity map has been reconstructed by using the proposed method and the BPA. Here, BPA is used to evaluate the efficiency and precision of the proposed method. Since the sampling interval of the transmitting array does not satisfy the Nyquist criterion, the echoed data are upsampled in the transmitting direction before using the proposed method. In addition, zero padding is utilized in the receiving direction so that the echoed data are uniformly spaced in the receiving direction and the sampling intervals in spatial wavenumber for transmitting and receiving arrays are the same.

The imaging results are given in Fig. 4. No windowing is used in imaging processing. Comparing the imaging results of the proposed method and BPA, it can be seen that the proposed method can accurately reconstruct the target as the BPA. The theoretical computational load and the actual time consumption are listed in Table II. The actual time consumption of the proposed method is 224.09 s, which is very short in comparison with 28074.47 s required by the BPA. The results show that the proposed method has high accuracy and efficiency.

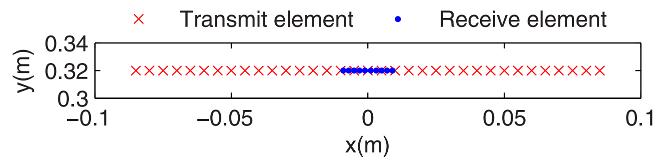

Fig. 5. Topology of the MIMO array.

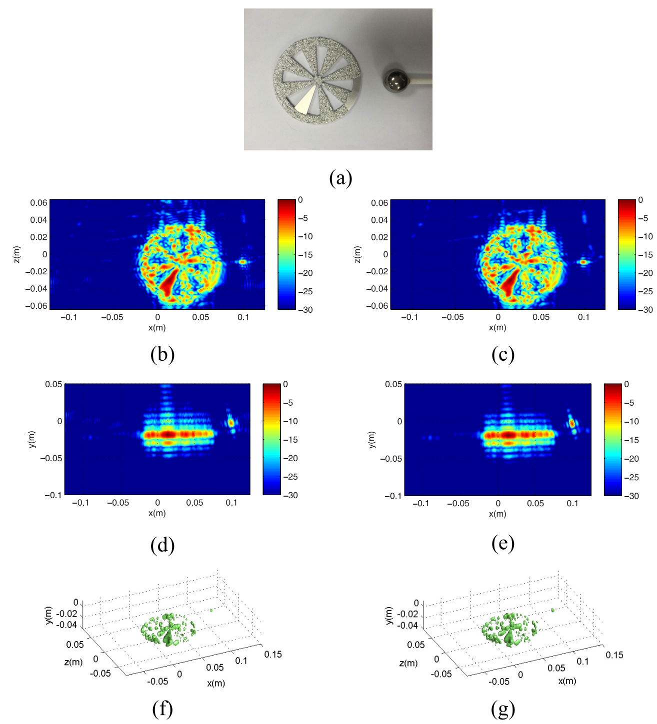

Fig. 6. Imaging results. (a) Photograph of the target. (b) Front view of image obtained by the proposed method. (c) Front view of image obtained by BPA. (d) Top view obtained by the proposed method. (e) Top view obtained by BPA. (f) Three-dimensional image obtained by the proposed method. (g) Three-dimensional image obtained by BPA. The top view is obtained by maximum projection of the 3-D image onto the *xy* plane.

B. Measurement Results

The proposed method has been validated experimentally by using nonideal scatterers. The photograph of the target is shown in Fig. 6(a). The target is composed of a Siemens star test pattern and a steel ball. The test pattern is approximately

TABLE II
COMPUTATIONAL LOAD AND IMAGING TIME

		Proposed method	BPA
Theoretical computational load (GFLOP)	Simulated data	97.57	4535.49
	Real data	36.99	1409.29
Actual time consumption(s)	Simulated data	224.09	28074.47
	Real data	187.52	8893.71

parallel to xz plane. The test pattern and the ball are not at the same range distance. The MIMO array consists of ten transmitting elements and 35 receiving elements; its topology is shown in Fig. 5. The array is located at 0.32 m from the target center. The transmitting element and receiving element are uniformly spaced. The spatial intervals of the transmitting array and the receiving array are 5 and 2 mm, respectively. Other parameters are the same as those in the simulation.

The imaging results of the proposed method and the BPA are given in Fig. 6. The echoed data are upsampled in the transmitting direction before using the proposed method. The theoretical computational load and the actual time consumption are listed in Table II. The results show that both methods can accurately reconstruct the target, whereas the proposed method greatly improves the imaging efficiency compared with BPA.

V. CONCLUSION

In this letter, an accurate and efficient 3-D imaging method for MIMO-SAR has been proposed. This method takes full advantage of the low computational load of FFT to improve imaging efficiency. For MIMO array not satisfying the Nyquist criterion, the echoed data need to be upsampled before FFT operation (or replicate the spectrum data after FFT operation) to obtain the complete spectrum. The overlapped spectrum caused by undersampling results in grating lobes in images, which provides a theoretical clue for suppressing grating lobes and will be explored in our further research. In addition, this method accurately compensates for the curvature of the wavefront in wavenumber domain; therefore, it has high accuracy and can be applied in near-field applications.

APPENDIX

To simplify the analysis, consider a 2-D imaging scene with an ideal point target. The locations of the target and the antenna are denoted by $(x,\ y)$ and $(u,\ y_0)$, respectively. For monostatic array, the received signal for a specific frequency can be expressed as

$$s(u) = e^{jk'\sqrt{(u-x)^2+(y_0-y)^2}}. \tag{A1}$$

By applying the method of stationary phase (MOSP), the inverse Fourier transform of $s(u)$ on u can be formulated as

$$\int s(u)e^{jk_u u}du \approx e^{jk_u x}e^{-j\sqrt{k'^2-k_u^2}(y_0-y)} \tag{A2}$$

where k_u represents the Fourier transform variable corresponding to u, and has the following relation according to the MOSP:

$$k_u = k'\frac{x-u}{\sqrt{(u-x)^2+(y_0-y)^2}}. \tag{A3}$$

By utilizing the Fourier transform relation, the following expression can be obtained from (A2):

$$s(u) = \int e^{jk_u x}e^{-j\sqrt{k'^2-k_u^2}(y_0-y)}e^{-jk_u u}dk_u. \tag{A4}$$

By substituting (A1) in (A4), the formula of the spherical wave decomposition is obtained, i.e.,

$$e^{jk'\sqrt{(u-x)^2+(y_0-y)^2}} = \int e^{-jk_u(u-x)}e^{-j\sqrt{k'^2-k_u^2}(y_0-y)}dk_u. \tag{A5}$$

Equation (A5) shows that the spherical wave can be decomposed into a superposition of plane wave components. By replacing the variables, (5)–(8) and (10) can be obtained.

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